

ISHAAN RAY GHATAK

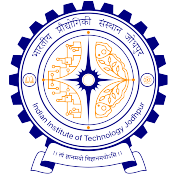
Indian Institute of Technology, Jodhpur

@ ishaanghatak2006@gmail.com

☎ (+91) 8758388899

in linkedin.com/in/ishaan-ghatak-99285b324/

github.com/ishaanghatak2006-tech



EXPERIENCE

Undergraduate Researcher

CLARITY Lab, IIT Jodhpur (Under Dr. Hardik Jain, Assistant Professor)

📅 Jan 2025 – Present

📍 Jodhpur, RJ

- Built [HandArM2](#), a gesture-controlled 3D model viewer using Python, MediaPipe, and Ursina Engine with on-demand model import via web scraping.
- Implemented Kalman filtering and a gesture state machine for smooth, robust real-time hand tracking, enabling intuitive interaction without keyboard or mouse input.

PROJECTS

CubiCode / LocalCodeJudge

Personal Project

- Built a self-hosted online judge REST API that accepts source code and custom test cases, securely compiles and executes submissions inside isolated Docker containers, and returns detailed verdicts and execution results.
- Developed a React-based coding workspace with structured test-case reporting, supporting completely offline code evaluation.
- Designed modular REST APIs with JWT authentication and MongoDB for problem management, submissions, user dashboards, and secure code execution workflows.

JaNET – Social Media Platform Analysis

Course Project – Data Structures & Algorithms

- Developed the backend analysis of a custom social media platform using graph-based user relationships and efficient data structures for feeds, follows, and user interactions.

Named Entity Recognition Analysis

Course Project – Pattern Recognition & Machine Learning

- Benchmarked HMM, MEMM, RNN, BiGRU, CNN, BiLSTM, and DistilBERT models for Named Entity Recognition.
- MEMM achieved 96.91% accuracy and the highest macro F1 score, while DistilBERT performed best among deep learning models.

Snail – Custom Linux Shell

Course Project – Software Engineering

- Built a custom Linux shell in C featuring reimplemented Unix commands, persistent history, trash/restore, interactive `rm -i`, and configurable shell themes.

Snake Game in C

Course Project – Introduction to Computer Science (Sem 1)

- Developed a terminal-based Snake game in C with score tracking and progressive difficulty.

EDUCATION

B.Tech, Computer Science & Engineering

Indian Institute of Technology, Jodhpur

📅 2023 – 2027 (Expected)

- CGPA: 8.84 (after Semester 4)

Class XII (CBSE)

Shanti Asiatic School, Ahmedabad

📅 2024

- Percentage: 91.4%

Class X (CBSE)

Shanti Asiatic School, Ahmedabad

📅 2022

- Percentage: 94.67%

TECHNICAL SKILLS

C++

C

Python

JavaScript

React

Node.js

Express.js

MongoDB

OpenCV

MediaPipe

LaTeX

Data Structures

Algorithms

Dynamic Programming

Graph Algorithms

Machine Learning

PROGRAMMING PROFILE

- 320+ problems solved on LeetCode
- Topics: Dynamic Programming, Graphs, Arrays, Strings, Linked Lists, Binary Search, Sliding Window
- Profile: leetcode.com/u/krakenwagen

LINKS

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</> LeetCode: krakenwagen